Multi-Criteria Decision Making: SMART and VIKOR Methods

Project Report

Ankara University
Department of Mathematics

Author: ***Şevket Kaan Alkır***

June 2022

Abstract

Many scientific techniques are used for decision making. While there are many informal and non-empirical methods applicable in specific areas, these do not generalize well to a broader context. However, using mathematical methods offers a universal approach for selecting the best alternative in any domain. This project introduces two frequently used Multi-Criteria Decision Making (MCDM) methods—SMART and VIKOR—and presents illustrative case studies.

Contents

1 Introduction to SMART Method 3
1.1 Key Definitions . . . 3
1.2 Introduction to the SMART Method . . . 3
1.3 SMART Method . . . 4
1.3.1 Attribute Rating . . . 4
1.3.2 Effective Weights of Alternatives . . . 4
1.3.3 Normalization of Effectiveness Values . . . 5
1.3.4 Final Performance of Alternatives . . . 5
1.4 Case Study: SMART Method . . . 5
1.5 Conclusion on SMART . . . 7

2 Introduction to the VIKOR Method 8
2.1 Definition of the LP-Metric . . . 8
2.2 VIKOR Method . . . 9
2.2.1 Best and Worst Values . . . 9
2.2.2 Computation of S and R Values . . . 9
2.2.3 VIKOR Index . . . 9
2.2.4 Final Ranking of Alternatives . . . 10
2.3 Case Study: VIKOR Method . . . 10
2.4 Conclusion on VIKOR . . . 11

1. Introduction to SMART Method

The Simple Multi-Attribute Rating Technique (SMART) was introduced by Winterfeldt and Edwards in 1986[1]. This method evaluates a limited number of alternatives using both quantitative and qualitative attributes. It is popular due to its ease of use and applicability across domains. The theoretical foundations for relative weighting in compensatory methods like SMART are further explored by Lootsma [2].

1.1 Key Definitions

- **Alternative:** Options to choose from.
- **Attribute:** Characteristics or features considered for evaluation.
- **Quantitative Attribute:** Measurable attributes.
- **Qualitative Attribute:** Non-measurable, descriptive attributes.
- **Positive Attribute:** Higher values are preferred (e.g., efficiency).
- **Negative Attribute:** Lower values are preferred (e.g., cost).

1.2 Introduction to the SMART Method

The SMART method, introduced by Winterfeldt and Edwards [1], was later refined as SMARTS and SMARTER by Edwards and Barron [3] to simplify multi-attribute decision-making. In this technique, a limited number of alternatives are evaluated based on a limited set of attributes. The primary goal is to assess both quantitative and qualitative attributes and rank the alternatives accordingly. This method is particularly valued for its ease of use. Key features of the SMART method include:

- It is classified as a compensatory method.
- It allows the use of both independent and dependent variables.
- Quantitative attributes can be converted into qualitative ones.

Initially, based on input from the decision-maker, a decision matrix is constructed as shown in Equation (1):

$$X = [x_{ij}], \quad i = 1, \ldots, m, \quad j = 1, \ldots, n \tag{1}$$

Here, x_{ij} represents the value of the j-th attribute for the i-th alternative. A standard 7-point linguistic scale is commonly used to rate attributes in SMART, as adopted in Haddara [4], providing a bridge between subjective evaluations and numerical scoring. The attributes are rated as shown in Table 1.

Table 1: Seven-point rating scale for attributes.

Weak	Fairly Weak	Normal	Fairly Good	Good	Very Good	Excellent
4	5	6	7	8	9	10

1.3 SMART Method

1.3.1 Attribute Rating

First, the minimum ($P_{\min}$) and maximum ($P_{\max}$) values for each attribute are defined by the decision-maker. The evaluation range is then divided into equally spaced subintervals as follows:

$$P_{\min},\ P_{\min} + e_0,\ P_{\min} + e_1,\ \ldots \tag{2}$$

To compute the intervals e_v, Equation (3) is used:

$$e_v - e_{v-1} = \varepsilon e_{v-1} \tag{3}$$

Where e_0 is the base step size between rating levels. Continuing this recurrence, we obtain:

$$e_v = (1+\varepsilon)e_{v-1} = (1+\varepsilon)^2 e_{v-2} = \cdots = (1+\varepsilon)^v e_0 \tag{4}$$

Eventually, the maximum value is expressed as:

$$P_{\max} = e_v + P_{\min} \tag{5}$$

1.3.2 Effective Weights of Alternatives

Let g_{ij} denote the effective weight of attribute C_j for alternative A_i, determined based on the decision-maker's judgment.

Qualitative attributes are rated according to Table 1. Quantitative attributes use Equation (1) and the value v representing the level of the attribute under consideration:

$$v = (\text{computed value based on scale}) \tag{6}$$

Here, v is the decision-maker's mapped score based on the linguistic scale in Table 1. For example, a 60% efficiency corresponds to a rating of 10, so $v = 6 \implies g_{ij} = 10$ for a positive attribute.

If the attribute is positive (i.e., higher values are better, e.g., product lifespan), then:

$$g_{ij} = v + 4 \tag{7}$$

If the attribute is negative (i.e., lower values are better, e.g., product price), then:

$$g_{ij} = 10 - v \tag{8}$$

1.3.3 Normalization of Effectiveness Values

The decision-maker is asked to assign a priority score to each attribute on the scale from 4 to 10, based on Table 1. To express this clearly, let us define:

- A_i: Alternatives, $i = 1, \ldots, m$
- C_j: Attributes, $j = 1, \ldots, n$
- h_j: Raw score assigned to attribute C_j
- w_j: Unnormalized weight of attribute C_j

The unnormalized weights are computed as:

$$w_j = \frac{h_j}{\sum_{j=1}^{n} h_j}, \quad j = 1, \ldots, n \tag{9}$$

1.3.4 Final Performance of Alternatives

The overall score for each alternative A_i is calculated by:

$$f_i = \sum_{j=1}^{n} w_j g_{ij}, \quad i = 1, \ldots, m \tag{10}$$

The alternative with the highest f_i is considered the best, while the remaining alternatives are ranked accordingly.

1.4 Case Study: SMART Method

A car engine manufacturing factory seeks to determine which of three engine types (A_1, A_2, A_3) it should produce, based on four attributes: Customer Price (C_1), Efficiency (C_2), Greenhouse Gas Emission (C_3), and Fuel Tank Capacity (C_4).

The ranges and desired directions for these attributes are:

- **Customer Price:** 20,000 TL – 40,000 TL (lower is better)
- **Efficiency:** 20% – 60% (higher is better)
- **Greenhouse Gas Emission:** 4.6 tons/year – 6.5 tons/year (lower is better)
- **Fuel Tank Capacity:** 50 liters – 150 liters (higher is better)

The decision-maker assigns importance levels to each attribute, as shown in Table 2. The attribute performance levels are listed in Table 3.

Table 2: Importance levels of attributes

Attribute	C_1	C_2	C_3	C_4
Rating	8	6	5	7

Table 3: Performance values of attributes (based on ratings)

Rating	Description	C_1	C_2	C_3	C_4
10	Excellent	20000	60	4.6	150
9		23000	58	4.8	125
8	Good	25000	50	5.0	110
7		30000	48	5.4	100
6	Medium	30300	35	5.9	70
5		35000	30	6.0	65
4	Poor	40000	20	6.5	50

Attribute Effectiveness Values

Using the decision-maker's ratings and Equation (6), effectiveness values g_{ij} are computed. For example, for the first alternative:

$$g_{11} = 6 \quad \text{(lower price is better)}$$
$$g_{12} = 9.807 \quad \text{(higher efficiency is better)}$$
$$g_{13} = 6.662 \quad \text{(lower emission is better)}$$
$$g_{14} = 9.678 \quad \text{(higher fuel capacity is better)}$$

The full matrix of effectiveness values is given in Table 4.

Table 4: Effectiveness values g_{ij} for each alternative

	A_1	A_2	A_3
C_1	6.000	5.000	4.415
C_2	9.807	9.321	5.584
C_3	6.662	5.078	9.807
C_4	9.678	8.848	6.678

Normalization of Attribute Weights

Using the raw importance ratings from Table 2 and Equation (9), we normalize the weights as follows:

- Customer Price (8): $w_1 = \frac{8}{8+6+5+7} = \frac{8}{26} \approx 0.390$

- Efficiency (6): $w_2 = \frac{6}{26} \approx 0.231$
- Emission (5): $w_3 = \frac{5}{26} \approx 0.192$
- Fuel Capacity (7): $w_4 = \frac{7}{26} \approx 0.269$

Rounded values used in calculations are listed in Table 5.

Table 5: Normalized attribute weights

Attribute	C_1	C_2	C_3	C_4
Weight	0.390	0.195	0.138	0.276

Final Ratings of Alternatives

Using Equation (10), we compute the final scores for each alternative:

$$f_1 = 0.390 \times 6 + 0.195 \times 9.807 + 0.138 \times 6.662 + 0.276 \times 9.678 = 7.842$$
$$f_2 = 0.390 \times 5 + 0.195 \times 9.321 + 0.138 \times 5.078 + 0.276 \times 8.848 = 6.910$$
$$f_3 = 0.390 \times 4.415 + 0.195 \times 5.584 + 0.138 \times 8.848 + 0.276 \times 6.678 = 5.874$$

Therefore, the ranking is: $A_1 > A_2 > A_3$, and the engine type A_1 is selected.

1.5 Conclusion on SMART

The SMART method, proposed by Winterfeldt and Edwards, stands out due to its ability to incorporate both qualitative and quantitative attributes without requiring their independence. This has led to its widespread development and application since its inception in 1986. As a compensatory approach, it is considered one of the most effective techniques for selecting the best alternative among multiple options. Moreover, the method's small number of steps makes the decision process efficient and fast. However, the compensatory nature of SMART may underperform in cases with high conflict between attributes, where a concession-based approach such as VIKOR might be more suitable

2. Introduction to the VIKOR Method

VIKOR was introduced by Opricovic [5] as a multicriteria optimization tool aimed at identifying compromise solutions in complex decision-making problems. It belongs to the family of compensatory models but operates as a concession-based method, aiming to select the alternative closest to an ideal solution within a subset.

Broadly, the technique consists of ranking alternatives, evaluating them based on conflicting attributes, and identifying a compromise solution that helps the decision-maker arrive at the final choice.

Since its inception, the VIKOR method has been applied in a variety of domains, such as analyzing logistics outsourcing[6], selecting airport locations[7], and to support post-disaster urban reconstruction planning [8]. The method's key features include:

- It is a compensatory decision-making method.
- Attributes are assumed to be independent.
- Quantitative attributes can be transformed into qualitative measures.

Based on input from the decision-maker, a decision matrix is constructed as in Equation (11):

$$X = [x_{ij}], \quad i = 1, \ldots, m, \quad j = 1, \ldots, n \tag{11}$$

Here, x_{ij} denotes the value of the j-th attribute for the i-th alternative. The decision-maker also provides the normalized attribute weights as follows:

$$[w_1, w_2, \ldots, w_n] \tag{12}$$

These are used in the standardization step (Equation 13) to ensure attributes are measured on comparable scales:

$$(\text{Standardized values}) = \ldots \tag{13}$$

2.1 Definition of the LP-Metric

The use of LP-metrics within VIKOR to measure proximity to the ideal solution is discussed in detail by Alinezhad and Esfandiari [9]. To find a solution closest to the optimal one, the VIKOR method utilizes an LP-metric approach (Equation 14):

$$L_p^i = \left[\sum_{j=1}^{n} w_j \left(\frac{f_j^* - f_{ij}}{f_j^* - f_j^-} \right)^p \right]^{1/p}, \quad i = 1, \ldots, m;\ 1 \leq p \leq \infty \tag{14}$$

Here, w_j is the weight of attribute j, and f_{ij} is the performance value of the i-th alternative on the j-th attribute. f_j^* and f_j^- denote the best and worst values, respectively, across all alternatives for attribute j.

For $p = 1$, the metric reduces to S_i (Equation 15):

$$S_i = \sum_{j=1}^{n} w_j \left(\frac{f_j^* - f_{ij}}{f_j^* - f_j^-} \right), \quad i = 1, \ldots, m \tag{15}$$

For $p = \infty$, the metric becomes R_i (Equation 16):

$$R_i = \max_j \left[w_j \left(\frac{f_j^* - f_{ij}}{f_j^* - f_j^-} \right) \right], \quad i = 1, \ldots, m \tag{16}$$

practice, the VIKOR method often uses $p = 1$ and $p = \infty$, corresponding to group utility (S_i) and individual regret (R_i) respectively.

2.2 VIKOR Method

2.2.1 Best and Worst Values

For each attribute $j = 1, \ldots, n$, the best (f_j^*) and worst (f_j^-) values among all alternatives are identified. For benefit-type (positive) attributes, the best values are computed using:

$$f_j^* = \max_i f_{ij}, \quad f_j^- = \min_i f_{ij} \tag{17}$$

For cost-type (negative) attributes, the inverse applies:

$$f_j^* = \min_i f_{ij}, \quad f_j^- = \max_i f_{ij} \tag{18}$$

These values are used to normalize performance levels and determine proximity to the ideal and anti-ideal solutions.

2.2.2 Computation of S and R Values

The values S_i and R_i are calculated using Equations 15 and 16, representing the group utility and the individual regret, respectively.

2.2.3 VIKOR Index

Wei and Lin [10] provide a formal derivation of the VIKOR index, which combines S and R values using a weight factor v. The VIKOR index Q_i is given by the formula:

$$Q_i = v \cdot \frac{S_i - S^*}{S^- - S^*} + (1 - v) \cdot \frac{R_i - R^*}{R^- - R^*} \tag{19}$$

where $S^* = \min_i S_i$, $S^- = \max_i S_i$, $R^* = \min_i R_i$, and $R^- = \max_i R_i$. The parameter $v \in [0, 1]$ represents the weight of the decision strategy. The parameter $v \in [0, 1]$ reflects the decision-maker's emphasis on majority satisfaction versus individual regret. Typically, $v = 0.5$ is used, where both are weighted equally.

2.2.4 Final Ranking of Alternatives

Alternatives are ranked based on their S, R, and Q values. The alternative with the lowest Q value is considered the most desirable. If the difference between the best and second-best alternatives is small, additional conditions may be required to ensure a stable compromise solution [8].

2.3 Case Study: VIKOR Method

A cargo company wishes to select the most suitable warehouse among three candidate locations (A_1, A_2, A_3) for storing shipments. The decision is based on two attributes, which the decision-maker considers equally important:

- Daily Storage Capacity (C_1): Higher is better
- Monthly Rent (C_2): Lower is better

The decision matrix is assumed to be as shown in Equation (20):

$$X = [x_{ij}], \quad i = 1, 2, 3; \quad j = 1, 2 \tag{20}$$

Best and Worst Values

The best and worst values for each attribute are:

- $f_1^* = 500$, $f_1^- = 233$ (Daily Capacity)
- $f_2^* = 8250$, $f_2^- = 10000$ (Monthly Rent)

Computation of S and R Values

Using Equation (15), the S_i values for each alternative are calculated as follows:

$$\begin{aligned} S_1 &= 0.5 \\ S_2 &= 0.5 \\ S_3 &= 0.296 \end{aligned}$$

Using Equation (16), the R_i values for each alternative are computed as:

$$\begin{aligned} R_1 &= 0.5 \\ R_2 &= 0.5 \\ R_3 &= 0.112 \end{aligned}$$

Then, we identify:

$$\begin{aligned} S^* &= 0.296, & S^- &= 0.5 \\ R^* &= 0.112, & R^- &= 0.5 \end{aligned}$$

Computation of VIKOR Index

Using Equation (19) and setting $v = 0.5$, we calculate the VIKOR index Q_i for each alternative:

$$
\begin{aligned}
Q_1 &= 0.5 \cdot \frac{0.5 - 0.296}{0.204} + 0.5 \cdot \frac{0.5 - 0.112}{0.388} = 0.5 + 0.5 = 1 \\
Q_2 &= Q_1 = 1 \\
Q_3 &= 0.5 \cdot \frac{0.296 - 0.296}{0.204} + 0.5 \cdot \frac{0.112 - 0.112}{0.388} = 0
\end{aligned}
$$

Final Ranking of Alternatives

Since $Q_3 < Q_1 = Q_2$, alternative A_3 is clearly the best choice based on the VIKOR index. The other two alternatives are equally less desirable.

$$Q_3 = 0 < Q_1 = Q_2 = 1 \quad \Rightarrow \quad \text{Select } A_3$$

2.4 Conclusion on VIKOR

The VIKOR method remains a valuable and popular decision-making tool due to its short procedural steps and the clarity it provides through the S, R, and Q indices. The final ranking of alternatives is determined based on essential input and a structured evaluation of diverse, compensatory criteria. Furthermore, modern software tools support the easy application of the VIKOR method in practical scenarios.

References

[1] von Winterfeldt, D., & Edwards, W. (1986). *Decision Analysis and Behavioral Research*. Cambridge University Press.

[2] Lootsma, F. A. (1996). A model for the relative importance of the criteria in the multiplicative AHP and SMART. *European Journal of Operational Research*, 94(3), 467–476.

[3] Edwards, W., & Barron, F. H. (1994). SMARTS and SMARTER: Improved simple methods for multi-attribute utility measurement. *Organizational Behavior and Human Decision Processes*, 60(3), 306–325.

[4] Haddara, M. (2014). ERP selection: The SMART way. *Procedia Technology*, 16, 394–403.

[5] Opricovic, S. (1998). *Multicriteria Optimization of Civil Engineering Systems*. Faculty of Civil Engineering.

[6] Kumar, R., Singh, H., & Dureja, J. S. (2012). An approach to analyze logistic outsourcing problem in medium-scale organization by CFPR and VIKOR. *Journal of Manufacturing Technology Management*, 23(7), 885–898.

[7] Sennaroglu, B., & Celebi, G. V. (2018). A military airport location selection by AHP integrated PROMETHEE and VIKOR methods. *Transportation Research Part D: Transport and Environment*, 59, 160–173.

[8] Opricovic, S., & Tzeng, G. H. (2002). Multicriteria planning of post-earthquake sustainable reconstruction. *Computer-Aided Civil and Infrastructure Engineering*, 17(3), 211–220.

[9] Alinezhad, A., & Esfandiari, N. (2012). Sensitivity analysis in the QUALIFLEX and VIKOR methods. *Journal of Optimization in Industrial Engineering*, 5(10), 29–34.

[10] Wei, J., & Lin, X. (2008). The multiple attribute decision-making VIKOR method and its application. In *4th International Conference on Wireless Communications, Networking and Mobile Computing (WiCOM'08)* (pp. 1–4). IEEE.